

Asymptotic estimated digestibility, a new indicator of black soldier fly (*Hermetia illucens*) conversion efficiency in relation to larval density

J.B. Guillaume^{1,2,3*} , S. Mezdour⁴ , F. Marion-Poll^{2,5} , C. Terrol³ and P. Schmidely¹ 

¹Université Paris-Saclay, INRAE, AgroParisTech, UMR Modélisation Systémique Appliquée aux Ruminants, 91120 Palaiseau, France; ²Laboratoire Évolution, Génomes, Comportement et Écologie, CNRS, IRD, Université Paris-Saclay, Institut Diversité, Ecologie et Evolution du Vivant (IDEEV), 91190 Gif-sur-Yvette, France; ³Agronutris, R&D Department, 31650 Saint-Orens de Gameville, France; ⁴Université Paris-Saclay, Sayfood, AgroParisTech, INRAE, 91120 Palaiseau, France; ⁵Université Paris-Saclay, AgroParisTech, 91120 Palaiseau, France; jeremy.guillaume@agroparistech.fr

Received: 22 July 2022 / Accepted: 28 November 2022

OPEN ACCESS 

RESEARCH ARTICLE

Abstract

Black soldier fly larvae (BSFL) are highly efficient at converting organic substrates into insect biomass suitable for livestock nutrition. BSFL conversion efficiency depends on the nature of their feed, and formulation approaches use macronutrient contents to reach a targeted performance. However, digestibility varies between ingredients and BSFL digestibility data is lacking due to methods unsuitable for insects living in their feed. This study proposes to work under a range of larval densities to reach total ingestion of the feed, and defines Estimated Digestibility (ED) as the difference between distributed feed and frass macronutrient weight, divided by macronutrient weight in distributed feed. Using the same amount of chicken feed, same feeding time and 0 to 29 larvae/cm², ED of dry mass (DM), starch, nitrogen, ether extract (EE), neutral detergent fibre, acid detergent fibre, acid detergent lignin, ash and energy were evaluated, along with survival rate, feed conversion ratio (FCR) and larval individual mass gain on a fresh and dry basis. ED of DM, starch, nitrogen, EE and energy increased with larval density following an asymptotic trend. ED of ash increased between 0 and 11.4 larvae/cm² but decreased for higher densities. Using a logistic model, asymptotic ED of DM (80.3±1.3%), starch (99.0±2.3%), nitrogen (78.6±1.1%), EE (95.3±1.5%), ash (58.4±1.0%) and energy (80.6±1.2%) were determined. No effect of areal density on ED of fibre fractions was detected. Survival rate and individual mass gain peaked at densities below 5 larvae/cm², FCR (dry to fresh) was improved with increasing density, while FCR (dry to dry) was optimised at 11.4 larvae/cm². Asymptotic ED is a new indicator corresponding to the maximal fraction of macronutrients potentially digestible by BSFL. It should be used to develop a diet formulation approach based on digestible rather than crude macronutrient contents. Further studies should focus on evaluating asymptotic ED in different ingredients.

Keywords: insect nutrition, formulation, macronutrient, alternative protein, Diptera

1. Introduction

Black soldier fly larvae (BSFL; *Hermetia illucens* (L.); Diptera: Stratiomyidae) can quickly convert organic side-streams and biowaste into valuable proteins and lipids (Barragan-Fonseca *et al.*, 2017) which can be used in animal nutrition (Bosch and Swanson, 2021; Dörper *et al.*, 2021) and have been increasingly produced and studied by academic and private stakeholders of the insects as feed sector. Both profitability and sustainability of BSFL production depend on their ability to convert feed

provided to the insects into larval biomass. Conversion efficiency varies namely with feed, rearing conditions and genetics (Greenwood *et al.*, 2021). Significant variability of zootechnical performances (development time, prepupal weight, larval protein and lipid contents, etc.) due to the type of feed and its macronutrient composition (mostly carbohydrates, proteins, lipids, fibres and ash) has been extensively documented (Gold *et al.*, 2018; Hopkins *et al.*, 2021). Barragan-Fonseca *et al.* (2021) used the Nutritional Geometry Framework to comprehensively map the impact of feed protein and carbohydrate content on BSFL

performance. This work laid the ground for a conceptual diet formulation approach that would allow targeting the appropriate macronutrient profile to reach the expected zootechnical performance of BSFL. However, BSFL fed diets formulated with the same protein and carbohydrate contents but different ingredients showed bioconversion rates of dry mass (DM) ranging from 14.5 to 31.8% (Gold *et al.*, 2020a). One main reason for this variability is the use of crude macronutrient fractions for the formulation instead of the more appropriate digestible macronutrient contents. Indeed the digestibility of a macronutrient can significantly vary from one ingredient to another, as extensively documented in conventional livestock species such as pigs (Stein *et al.*, 2016) or poultry (Zaefarian *et al.*, 2015). Still, data on digestibility by BSFL are lacking.

Apparent digestibility in conventional livestock species is defined on a DM basis for any macronutrient as the amount of macronutrient in ingesta subtracted by macronutrient in faeces, divided by macronutrient in ingesta (Barret *et al.*, 2005). Its calculation requires: (1) estimating the amount of feed ingested; and (2) collecting pure faeces (not contaminated with feed, urine, bedding, etc.) to perform macronutrient content analyses. Ingested feed is usually either determined gravimetrically by comparing the weights of distributed – all the feed given to the animal – and refused feed – non-ingested feed left at the end of the feeding period – or via an undigestible tracer added to the feed and quantified in excreta, such as chromic oxide (McGinnis and Kasting, 1964). However, BSFL feed in aggregation on moist substrates that also constitute their living environment along with various microbes, and excreta cannot be separated from non-ingested feed, exuviae abandoned during moulting and potential dead individuals (Seyedalmoosavi *et al.*, 2022). All these components are mixed together and indistinguishable in a powdery to sticky compost-like material called ‘frass’; therefore, any undigestible tracer would ultimately end up in the frass and collecting reliable data to calculate Apparent Digestibility in BSFL is challenging. A possible partial solution is to attach an excreta collection device to the bottom end of a larva. Still, collected amounts of pure excreta are tiny and this method is impractical on many larvae (Gold *et al.*, 2020b). Furthermore, unlike mammals that produce faeces and urine separately, insects’ metabolic end products are released in the hindgut via Malpighian tubules (Weihrach and O'Donnell, 2021). Hence their excreta consists of a mix of faeces (undigested feed material) and urine (metabolic end products). Accordingly, Waldbauer (1968) highlighted the difference between Approximate Digestibility, which is calculated based on macronutrient weight in ingesta and in excreta, and Apparent Digestibility, which is similar but requires subtracting the weight of macronutrients in urine from total excreta to obtain the weight of macronutrients in faeces and thus be consistent with the definition of apparent

digestibility in conventional livestock. This refinement is suitable when insects’ excreta can be separated from non-ingested feed, which is not the case in BSFL. Yet, some authors analysed frass to gain insight into BSFL nutrition but imprecisely referred to approximate digestibility (Klammsteiner *et al.*, 2021; Permana and Ramadhani Eka Putra, 2018). This misuse is due to confusion between distributed and ingested feed and an unclear definition of BSFL excreta and frass (Bosch *et al.*, 2020). In the present paper, we consider excreta as a mixture of faeces and urine, and we use the official definition of frass given by the European Commission: ‘the mixture of insect excrements with parts of dead insects and feeding substrate’ (European Commission, 2021). Hence, we propose a new definition for a digestibility approximation specific to insects growing in their feed: estimated digestibility (ED) is calculated by dividing the difference between distributed feed and frass macronutrient weight by the amount of macronutrient in distributed feed. For insects feeding on moist and microbe-rich substrates like BSFL, ED should be considered as the overall digestion efficiency of insects and the substrate’s microbial community (Schreven *et al.*, 2021).

DM content of frass is often measured in studies focusing on waste management with BSFL to quantify ‘waste reduction’ (sometimes called ‘overall degradation’). These terms have the same definition as ED of DM but are imprecise and unsuitable in the field of animal nutrition. ED of DM ranges from 9 to 85% depending on the type of feed but also rearing conditions, such as feed moisture content, microbes in the substrate, feeding duration or larval density (Lalander *et al.*, 2019; Rehman *et al.*, 2017). The latter’s effect on ED has only been indirectly investigated, with results suggesting that ED is positively associated with larval density (Brits, 2017; Permana and Ramadhani Eka Putra, 2018). This is supported by the hypothesis that a higher number of BSFL is more likely to consume a given amount of feed entirely. It can be assumed that a high density would ultimately result in reaching a maximal ED, corresponding to the full potential of the feed that BSFL and the microbial community can digest. However, this has never been documented. In contrast to ED of DM, values of ED of starch, protein, lipid, fibre and ash are scarce to non-existent.

Consequently, this study proposes a method to assess the maximal ED of a feed for the BSFL conversion process. Null to extremely high larval densities were used, and an extensive analysis of remaining macronutrients in frass was performed to challenge the impact of density on the ED of each macronutrient. This approach aims to consider the maximal ED of a macronutrient as the digestibility coefficient of those specific macronutrient and ingredient when developing a BSFL diet formulation tool.

2. Materials and methods

Insects and feed used

BSFL were obtained from Agronutris (Toulouse, France). Before the trial, neonate larvae were fed chicken feed (CF) (Maisadour, Sud-Ouest Aliment VT100 Poulet Jaune Démarrage, Haut-Mauco, France) rehydrated to $64 \pm 0.5\%$ moisture. On a DM basis, the CF contained $42.1 \pm 1.9\%$ starch (mean and standard-deviation on four samples), $3.4 \pm 0.2\%$ nitrogen (four samples), $2.8 \pm 0.1\%$ ether extract (EE; two samples), $6.1 \pm 0.2\%$ ash (two samples), $14.4 \pm 0.9\%$ neutral detergent fibre (NDF; five samples), $5.7 \pm 0.7\%$ acid detergent fibre (ADF; five samples), $1.3 \pm 0.9\%$ acid detergent lignin (ADL; five samples) and had a gross energy content of 15.8 ± 0.5 kJ/g DM. Pre-experimental BSFL were kept at around 28°C and 55% relative humidity (RH) for about seven days. After shipping to the laboratory, they were successively sieved with 2.4 and 1.0 mm diameter sieves to get rid of most of the frass and standardise their size by removing over- and under-sized larvae. The larvae were then gently stirred by hand and three samples of about 1 g were taken and counted manually to measure the average fresh mass (FM) of a starter larva (4.9 ± 0.2 mg) and to calculate the overall mass of larvae to use for each experimental density.

Experimental rearing conditions

Experiments were conducted in $16.5 \times 10.6 \times 7$ cm rectangular plastic containers (Pro-Jet BG110, Saint-Ouen-l'Aumône, France) individually placed in a knotted light pantyhose and sealed with a lid with an opening of 16×10 cm to prevent larvae from escaping while allowing air circulation. The feed used was CF rehydrated to $68 \pm 0.5\%$ moisture using hot tap water and mixed thoroughly (Electrolux Professional Bermixer Pro 450 W, Saint-Denis, France; maximum power for 2:30 min) right before distribution. Each container received a unique input of feed for the whole experimental period, consisting of 420 ± 1 FM g of CF, i.e. 134 DM g, and a specified amount of larvae, covering an extensive range of larval density (Table 1). These experimental conditions were established based on preliminary tests to ensure presumed total feed consumption and sufficient frass DM collection for analyses. Two trials were performed in the same conditions, with two and three replicas per density respectively and slight changes in the investigated larval densities.

Containers were placed in a climatic chamber at $28 \pm 0.5^\circ\text{C}$ and $75 \pm 2\%$ RH, with a L12:D12 photoperiod (Trial 1: Xylem Analytics WTW TS 1006-i, Weilheim, Germany; Trial 2: Percival Scientific DR-36VL, Perry, GA, USA). Larvae were kept in their container and allowed to feed for seven days. Containers were relocated eight times to minimise effects related to possible climatic heterogeneity within the chamber. No additional feed input, sampling or mixing was performed during the trial period.

Table 1. Experimental larval densities on a surface and dry feed mass basis.

Number of larvae per container	Areal density (larvae/cm ²)	Mass density (larvae/kg dry mass)	Feed availability (mg dry mass/larva)	Trial
0	0	0	-	1 & 2
400	2.3	2,976	336	1 & 2
600	3.4	4,464	224	1
700	4.0	5,208	192	2
800	4.6	5,952	168	1
1000	5.7	7,440	134	1 & 2
1,500	8.6	11,161	90	1 & 2
2,000	11.4	14,881	67	1 & 2
3,000	17.1	22,321	45	1 & 2
5,000	28.6	37,202	27	1 & 2

Sampling and analyses

At the end of the seven-day feeding period, all containers were removed from the chamber and weighed. Larvae were manually separated from the frass. Depending on the stickiness of the frass, they were either sieved or removed with forceps. They were rinsed with tap water in a fine sieve, dried with paper towel and weighed collectively. The FM of frass was calculated as the final weight of the container before removing the larvae, subtracted by the weight of the cleaned larvae and the weight of the empty container.

Frass, including material sticking to the inner side of the container, was manually collected and mixed with a spatula. DM content was determined on a 4 g sub-sample using a thermobalance set at 120°C (Radwag MA 50.R, Radom, Poland). The chemical composition of frass and CF was analysed on oven-dried (Memmert UE500, Schwabach, Germany; 60°C until constant weight) and ground (Retsch ZM 1000, Haan, Germany; 1 mm grid) samples. Starch content was determined by colourimetry (Roche Cobas Mira S, Basel, Switzerland; Randox GLUC-PAP Kit, Crumlin, UK) after enzymatic hydrolysis (Merck, 10115-5G-F, Darmstadt, Germany) according to Faisant *et al.* (1995); nitrogen content was determined using the Dumas method (Elementar EA Vario Micro Cube, Langenselbold, Germany); EE was determined with the Soxhlet method using diethyl ether (Tecator Soxtec System HT6, Höganäs, Sweden); ash content was determined gravimetrically by complete combustion at 550°C for 8 h (Nabertherm Furnace, Lilienthal, Germany); NDF, ADF and ADL were determined using the filter bag technique (Ankom 220 Fibre Analyzer, Macedon, NY, USA) according to the Van Soest method, after treatment with acetone, α -amylase and pronase to prevent interactions with excess lipids, starch

and proteins respectively (Godin *et al.*, 2010). The gross energy of each sample was calculated using a coefficient of 17.17 kJ/g for starch and NDF, 23.45 kJ/g for proteins (defined as $N \times 6.25$) and 38.94 kJ/g for EE (Vaissaire, 2018). ED of DM, starch, nitrogen, EE, ash, NDF, ADF, ADL and energy was calculated for each container using Equation 1, where A stands for the considered macronutrient:

$$\text{Estimated Digestibility of A} = \frac{\text{Weight of A in distributed feed} - \text{Weight of A in frass}}{\text{Weight of A in distributed feed}} \times 100 \quad (1)$$

For each container, a sample of 30 larvae was weighed on a precision balance (Sartorius AZ153, Göttingen, Germany, ± 0.001 g) to calculate the mean individual FM gain and estimate the final number of larvae. The survival rate was calculated as the ratio between final and initial numbers of BSFL. Feed conversion ratio (FCR dry to fresh) was calculated as the ratio between DM of feed distributed and total larval FM gain. Larvae were killed by freezing at -20°C . Due to improper sample management, no further analysis was performed on Trial 1 larvae. Larvae from Trial 2 were oven-dried at 60°C until constant weight to calculate their DM content, mean individual DM gain and FCR on a dry basis (FCR dry to dry).

Statistical analysis

Collected data was analysed using R (version 4.1.2., R Core Team, 2021). As there was no difference of treatment between trials, except for a mean difference of recorded temperature of 1.0°C (27.1°C for Trial 1, 28.1°C for Trial 2), all data was pooled together. Considering the pattern of scatter plots of ED of DM, starch, nitrogen, EE, ash and energy, the effect of areal density was modelled for each variable using the nonlinear logistic model 'logit' from the 'drc' package. The equation of the model is Equation 2, with $-1/b$ the slope for $X=0$, c the lower asymptote, d the upper asymptote and e the inflexion point:

$$Y = c + \frac{d-c}{1+\exp(b(X-e))} \quad (2)$$

Similarly, the impact of areal density on FCR (dry to fresh) and FCR (dry to dry) was modelled using an exponential model with the 'nls' function from the 'stats' package. This model is given in Equation 3 where b is the decreasing rate, c the intercept for $X=0$ and d the asymptote.

$$Y = d + (c - d) \exp(-b * X) \quad (3)$$

These models were used to estimate the value of the asymptote and the areal densities required to reach 95 and 99% of the asymptotic performance (AD95 and AD99, respectively).

3. Results

Estimated digestibility

Figure 1 presents values of ED for each areal density. Increasing areal density between 0 and 11.4 larvae/ cm^2 positively affected ED of DM (from 3.0 ± 2.0 to $79.7 \pm 1.8\%$), starch (from 24.6 ± 7.8 to $98.7 \pm 1.2\%$), nitrogen (from -12.3 ± 3.4 to $78.4 \pm 2.0\%$), EE (from 9.9 ± 15.3 to $94.8 \pm 1.4\%$), ash (from -0.6 ± 1.6 to $60.8 \pm 4.4\%$) and energy (from 11.0 ± 2.3 to $80.3 \pm 1.2\%$). Higher densities showed similar ED values, resulting in an asymptotic trend, although ED of ash slightly decreased between 11.4 and 28.6 larvae/ cm^2 from $60.8 \pm 4.4\%$ down to $54.0 \pm 4.9\%$. Asymptotic ED determined by the logistic models (parameter d) were 80.3, 99.0, 78.6, 95.3, 58.4 and 80.6% for DM, starch, nitrogen, EE, ash and energy respectively (Table 2).

No effect of areal density on ED of NDF, ADF and ADL could be detected (Figure 1G, H, I). Mean ED of these fractions was $22.6 \pm 5.9\%$, $6.6 \pm 6.1\%$ and $11.8 \pm 15.6\%$, respectively.

Life history traits and conversion efficiency

Figure 2 presents the results of life history traits and conversion efficiency. Mean survival rate ranged between 100.2 ± 2.5 and $85.4 \pm 3.8\%$ and declined with increasing areal density. Values above 100% are a computational artefact due to a lack of accuracy in estimating the initial and final numbers of BSFL. FCR (dry to fresh) was continuously improved by increasing areal density until reaching a minimum level (0.73 ± 0.04), while FCR (dry to dry) only decreased between 2.3 and 11.4 larvae/ cm^2 (from 7.30 ± 0.60 to 3.02 ± 0.07) and worsened slightly at higher densities. Both were modelled with an exponential equation: asymptotic value, AD95 and AD99 were 0.78, 8.6 larvae/ cm^2 and 11.7 larvae/ cm^2 for FCR (dry to fresh) and 3.15, 5.9 larvae/ cm^2 and 7.5 larvae/ cm^2 for FCR (dry to dry) respectively (Table 2). Larval DM content, individual FM gain and individual DM gain reached their maximal value at 4.0 larvae/ cm^2 ($30.3 \pm 1.4\%$, 182.7 ± 13.4 mg and 55.4 ± 6.1 mg, respectively) and decreased with rising areal density.

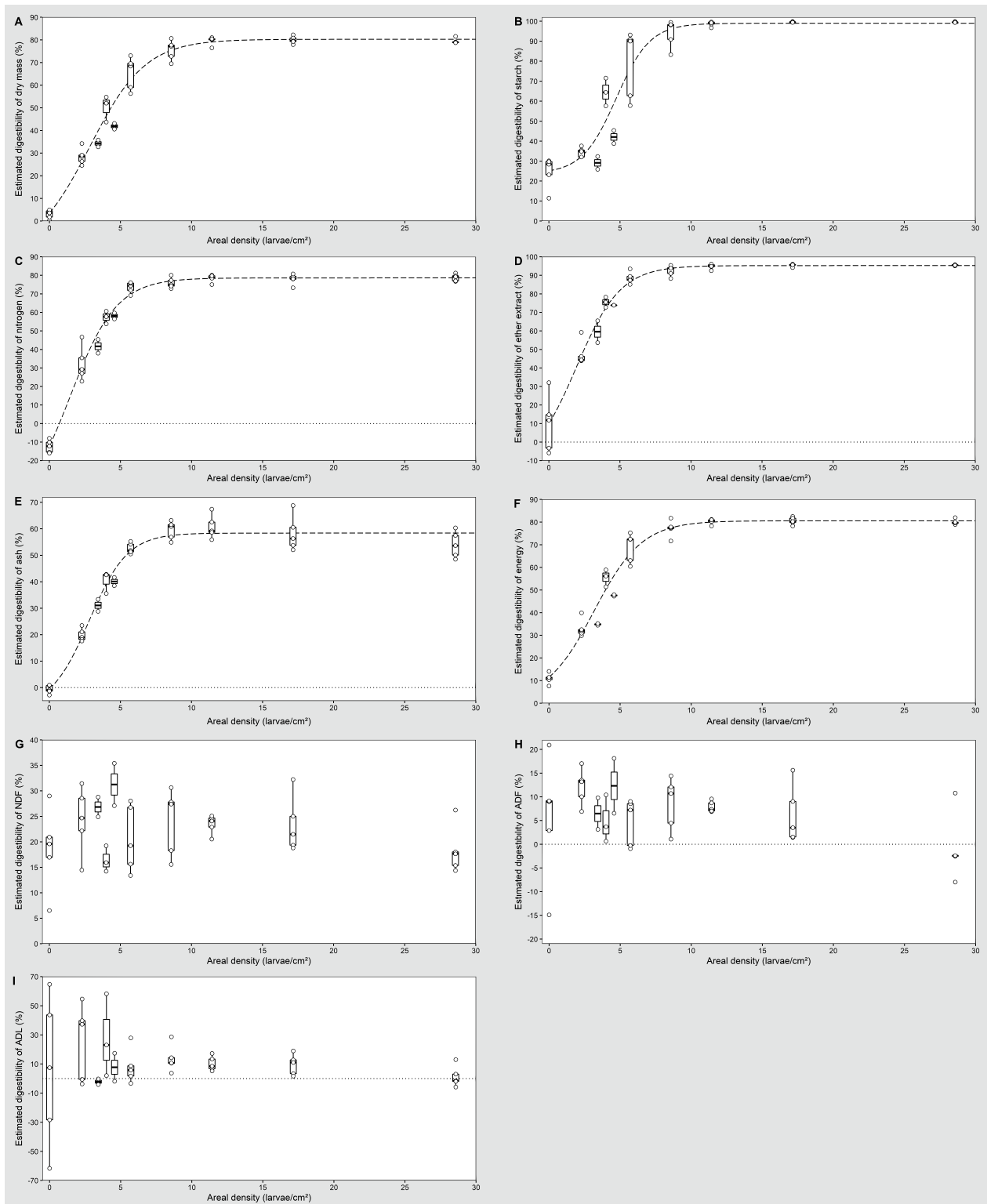


Figure 1. (A) Estimated digestibility of dry mass, (B) starch, (C) nitrogen, (D) ether extract, (E) ash, (F) energy, (G) neutral detergent fibre (NDF), (H) acid detergent fibre (ADF) and (I) acid detergent lignin (ADL) in black soldier fly larvae fed chicken feed and reared in seven days at varying densities.

Table 2. Root mean square prediction error (RMSPE), Bayesian information criterion (BIC) and parameters (intercept and standard error) of the models used to describe the effect of areal density on studied variables, and minimal estimated areal density to reach 95 and 99% of the asymptotic value (AD95 and AD99, respectively).¹

Variable	Type of regression	RMSPE	BIC	Model parameters				AD95 (larvae/cm ²)	AD99 (larvae/cm ²)
				b	c	d	e		
ED of dry mass	Logistic	4.532	264.8	-0.509 (0.07) ***	-14.507 (9.076)	80.252 (1.268) ***	2.839 (0.46) ***	9.0	12.2
ED of starch	Logistic	9.021	322.6	-0.839 (0.213) ***	23.778 (5.149) ***	98.975 (2.344) ***	4.731 (0.287) ***	7.9	9.9
ED of nitrogen	Logistic	4.394	262.2	-0.611 (0.081) ***	-49.254 (17.129) **	78.601 (1.109) ***	1.461 (0.521) **	7.1	9.8
ED of ether extract	Logistic	5.871	286.6	-0.644 (0.115) ***	-14.538 (15.274)	95.259 (1.486) ***	1.93 (0.544) **	6.7	9.3
ED of ash	Logistic	4.090	256.2	-0.742 (0.099) ***	-7.784 (4.613)	58.386 (0.993) ***	2.801 (0.272) ***	7.0	9.2
ED of energy	Logistic	4.420	262.7	-0.591 (0.083) ***	0.956 (6.233)	80.576 (1.179) ***	3.171 (0.348) ***	8.1	10.9
FCR (dry to fresh)	Exponential	0.074	-73.7	0.558 (0.036) ***	5.689 (0.437) ***	0.782 (0.018) ***	-	8.6	11.7
FCR (dry to dry)	Exponential	2.269	19.2	0.913 (0.129) ***	36.742 (9.97) **	3.148 (0.087) ***	-	5.9	7.5

¹ Level of significance: $P < 0.001$: ***; $P < 0.01$: **; $P < 0.05$: *. ED = estimated digestibility; FCR = feed conversion ratio.

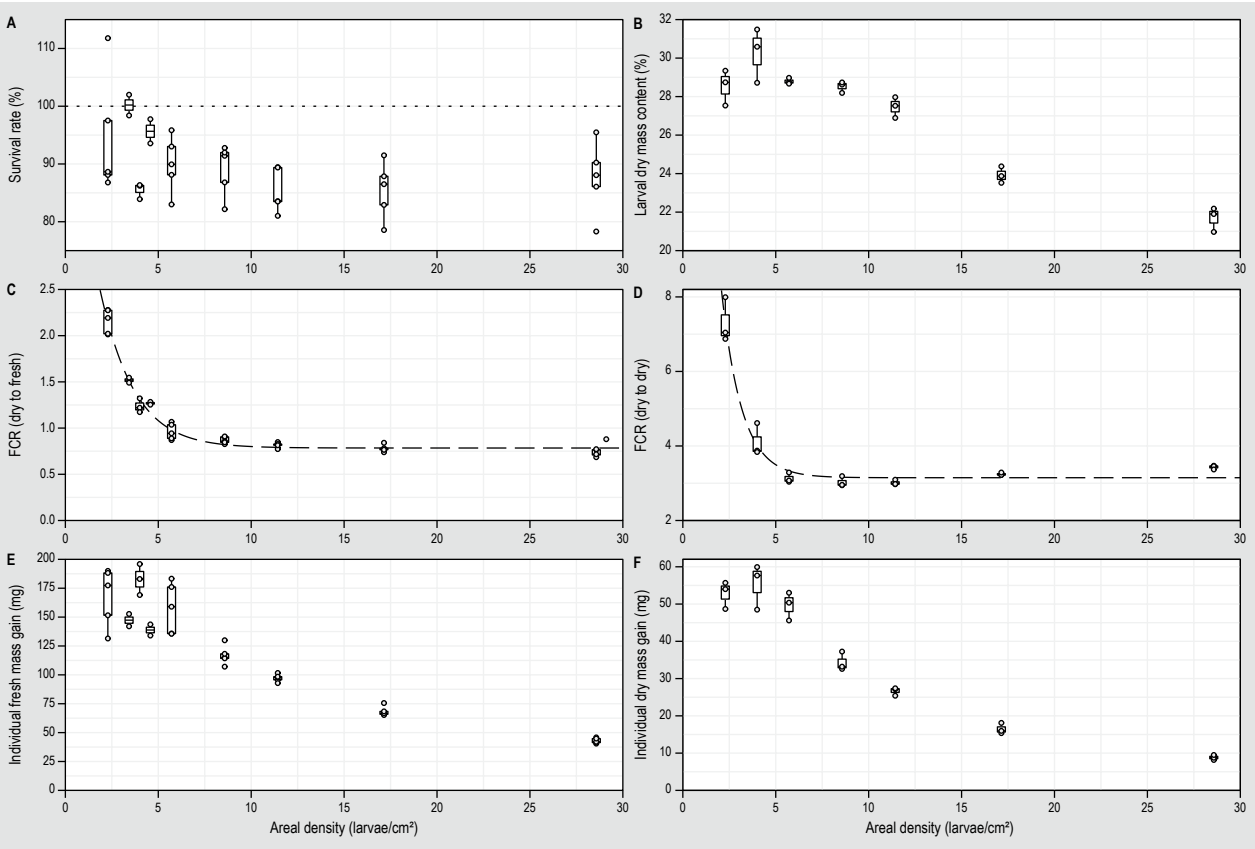


Figure 2. (A) Survival rate, (B) larval dry mass content, (C) feed conversion ratio (FCR) on a fresh basis, (D) FCR on a dry basis, (E) individual fresh mass gain and (F) individual dry mass gain of black soldier fly larvae fed chicken feed and reared in seven days at varying densities.

4. Discussion

This study proposed a method based on frass analyses and variation of larval density to assess maximal ED in BSFL conversion systems. The results showed that the ED of DM, starch, nitrogen, EE, ash and energy is positively affected by areal density, following a sigmoidal pattern that can be modelled with a logistic curve. High areal densities allowed more BSFL to collectively feed on the same amount of CF, leading to higher ED values, up to an asymptote that indicates the maximal fraction of feed

BSFL and associated microbes can potentially digest. Undigestible macronutrients remaining in the frass were not biologically available for BSFL either because they are physically embedded in a resistant structure (such as proteins in a vegetal cell wall) or because larvae do not have the appropriate enzymes. In CF, BSFL and the associated microbial community were able to digest up to 80.3, 99.0, 78.6, 95.3, 58.4 and 80.6% of DM, starch, nitrogen, EE, ash and energy respectively. ED of DM in CF in the literature ranged between 49.9 and 84.8% (Table 3). Brits (2017) explored the effect of density on ED of DM in kitchen

Table 3. Comparison of estimated digestibility in black soldier fly larvae (BSFL) for different types of feed and rearing conditions.¹

Type of feed	Number of BSFL	FM of diet (g)	Moisture content (%)	Larval density			Feeding time (day)	Number of feed supplies	Estimated digestibility											Reference ²
				Areal density (larvae/cm ²)	Mass density (larvae/kg DM)	Feed availability (mg DM/larva)			DM (%)	Carbon (%)	Nitrogen (%)	Starch (%)	Lipids (%)	Energy (%)	Phosphorus (%)	Potassium (%)	Cellulose (%)	Hemicellulose (%)	Lignin (%)	
Chicken feed	600	1,300	61.5	-	1,199	834	15	1	70.0	-	-	-	-	-	-	-	-	-	-	a
	200	310.7	58.8	0.6	1,563	640	-	Several	84.8	-	-	-	-	-	-	-	-	-	-	b
	80	48.0	60.0	1.8	4,167	240	9	3	67.7	-	-	-	-	-	-	-	-	-	-	c
	500	400	70.0	2.4	4,167	240	9	4	49.9	-	-	-	-	-	-	-	-	-	-	d
	80	41.0	60.0	2.8	4,878	205	9	1	54.7	-	-	-	-	-	-	-	-	-	-	e
	80	32.8	60.0	2.8	6,107	164	9	1	58.5	-	-	-	-	-	-	-	-	-	-	e
Kitchen waste	330	990	75.1	3.4	1,340	746	15	1	25.8	-	-	-	-	-	-	-	-	-	-	f
	440	990	75.1	4.5	1,786	560	15	1	32.9	-	-	-	-	-	-	-	-	-	-	f
	528	990	75.1	5.4	2,144	467	15	1	35.5	-	-	-	-	-	-	-	-	-	-	f
	660	990	75.1	6.7	2,680	373	15	1	35.0	-	-	-	-	-	-	-	-	-	-	f
	880	990	75.1	8.9	3,573	280	15	1	33.8	-	-	-	-	-	-	-	-	-	-	f
	1,320	990	75.1	13.4	5,359	187	15	1	34.2	-	-	-	-	-	-	-	-	-	-	f
CM	1000	1000	78.4	-	4,630	216	24.3	1	25.8	52.0	32.0	-	-	-	38.7	-	51.0	51.7	35.0	g
80% CM + 20% SBR	1000	1000	79.2	-	4,808	208	22.5	1	32.4	60.0	34.7	-	-	-	41.3	-	51.3	43.3	25.0	g
60% CM + 40% SBR	1000	1000	79.7	-	4,926	203	20.6	1	46.5	62.7	48.0	-	-	-	48.0	-	55.0	51.7	32.3	g
40% CM + 60% SBR	1000	1000	80.7	-	5,181	193	21.1	1	56.6	65.3	60.0	-	-	-	50.7	-	64.7	63.3	36.7	g
20% CM + 80% SBR	1000	1000	81.5	-	5,405	185	21.1	1	68.3	72.0	72.0	-	-	-	58.7	-	69.7	71.3	33.3	g
SBR	1000	1000	82.3	-	5,650	177	19.4	1	72.4	70.7	73.3	-	-	-	68.0	-	68.3	81.7	39.3	g
40% CM + 60% SBR	1000	1000	76.1	-	4,179	239	20.6	1	42.0	-	54.0	-	47.7	-	-	-	49.6	55.1	24.0	h
Pig manure	22,500	18,830	76.1	4.5	5,000	200	9	1	22.0	27.0	47.0%	-	-	30.0	9.0	6.0	-	-	-	i
Artificial diet	10,000	4,000	71.7	6.7	8,834	113	7	1	42.0	45.8	37.9%	55.8	72.9	46.8	27.5	12.6	-	-	-	j

¹ CM = cattle manure; DM = dry mass; FM = fresh mass; SBR = soybean curd residue.

² a = Gobbi *et al.* (2013); b = Lalander *et al.* (2019); c = Gold *et al.* (2020b); d = Broeckx *et al.* (2021); e = Cohn *et al.* (2022); f = Brits (2017); g = Rehman *et al.* (2017); h = Rehman *et al.* (2019); i = Parodi *et al.* (2021); j = adapted from Parodi *et al.* (2020).

waste and, although the experimental set-up did not allow for a clear asymptotic trend, a maximal value of 35.5% was obtained. This result suggests CF to be more digestible than kitchen waste. Values of ED of starch, nitrogen, EE, ash and energy in CF are unavailable in the literature, but some have been reported in other types of feed. ED of nitrogen ranged from 27% (in pig manure) to 72.4% (in soybean curd residue). In a mixture of yeast, wheat and potato by-products, ED of starch, nitrogen, lipids and energy were 55.8, 37.9, 72.9 and 46.8%, respectively (Table 3). Values of ED from the present study were the highest reported for these macronutrients. However, these results should be compared with caution due to a lack of standardisation in experimental conditions between studies, namely in terms of larval density, feeding time, feeding rate and genetics of the insects used (Bosch *et al.*, 2020; Zhou *et al.*, 2013).

Asymptotic ED in BSFL can also be compared to Apparent Digestibility and Approximate Digestibility in other insect species. Both vary significantly with feed, insect species, rearing conditions and larval instar. In mealworm larvae (*Tenebrio molitor*), wheat bran has an Approximate Digestibility of DM of 46% and an Apparent Digestibility of nitrogen, starch and lipids of 54, 59 and 73%, respectively (Waldbauer, 1968). The Apparent Digestibility of wheat in growing pigs and ruminants is 90 and 88% for organic matter, 84 and 70% for nitrogen, 24 and 63% for lipids, and 88 and 86% for energy, respectively (Sauvant *et al.*, 2002). Hence BSFL show strong potential to digest high-quality feed such as CF, comparable to what can be observed in conventional livestock species.

It can be noticed that mean ED of starch slightly decreased between 4.0 and 4.6 larvae/cm² (from 64.5±6.9 to 42.1±4.6%) and the same phenomenon can also be observed to a lesser extent in ED of other fractions. This is likely due to the way data was pooled, as the 4.0 and 4.6 larvae/cm² treatments were only present in Trial 2 and Trial 1 respectively. It can be assumed that bioconversion process was slightly improved in Trial 2 where the mean temperature recorded in the climatic chamber was 1.0 °C higher than in Trial 1. However, ED values above 11.4 larvae/cm² were highly consistent, which suggests that asymptotic ED is a reliable indicator of the digestibility of substrates in BSFL rearing, probably weakly affected by temperature.

In contrast with other macronutrients, ED of NDF, ADF and ADL were not affected by areal density. Mean ED values (22.6±5.9%, 6.6±6.1% and 11.8±15.6%, respectively) were below those of other fractions, suggesting that BSFL are inefficient at digesting fibres. These results are consistent with reported digestive activity in BSFL, which are high for amylase, lipase, and protease but low for cellulase (Intayung *et al.*, 2021; Sun *et al.*, 2022). This also concurs with BSFL developing well on substrates rich in starch and proteins and having low growth performance on fibre-rich substrates

(Gold *et al.*, 2018). However, these very low ED of fibres are inconsistent with results obtained on soybean curd residue, cattle manure and mixtures of both, where ED varied between 49.6 and 69.7% for cellulose, 43.3 and 81.7% for hemicellulose, 24.0 and 39.3% for lignin (Rehman *et al.*, 2017). This difference could indicate that fibre fractions are less digestible in CF than in cattle manure and soybean curd residue. However, low fibre ED from the present study could also result from an inappropriate analysis method. Indeed, it has been reported that the Van Soest method was not specific to vegetal fibre fractions and that ADF could be used to estimate chitin content in shellfish (Stelmock *et al.*, 1985). Although insect and shellfish chitin have structural differences (Brigode *et al.*, 2020), it can be assumed that at least part of the chitin from dead BSFL and exuviae in frass was quantified by the Van Soest method along with plant fibres. It remains unclear whether the Van Soest method is appropriate for estimating residual plant fibre content in frass. Other analysis methods should be explored, such as a pre-treatment of frass samples with chitinase. As chitin contains nitrogen, measuring frass chitin content would also allow a more accurate nitrogen ED determination.

With 0 larvae/cm², ED was positive for DM (3.0±2.0%), starch (24.6±7.8%), EE (9.9±15.3%) and energy (11.0±2.3%). This is likely due to the activity of microbiota in the substrate that partly digested it to produce microbial biomass. These values should not be considered as a basal microbial activity and used to arithmetically isolate digestion by BSFL and by microbes, as larval activity modifies the microbial community and both affect each other in a process called 'co-conversion' (Cifuentes *et al.*, 2020; Schreven *et al.*, 2022), with BSFL consuming and digesting microbial biomass (Gold *et al.*, 2018). On the contrary, 0 larvae/cm² density resulted in negative ED for nitrogen (-12.3±3.4%) and ash (-0.7±1.6%), which indicated a negative balance between macronutrients removed from and added to the substrate during the trial. Negative ED of nitrogen could result from diazotrophs in the feed that transformed atmospheric dinitrogen into microbial proteins or ammonia released in the substrate (Chan *et al.*, 1994). Without BSFL, the substrate kept a high moisture content: therefore, it can also be hypothesised that it absorbed some of the gaseous and hydrophilic ammonia produced by BSFL and microbes of other containers in the same climatic chamber. Conversely, the pH increase, heat production and larval movement caused by BSFL digestion process resulted in nitrogen loss from the system via ammonia volatilisation (Parodi *et al.*, 2021), hence overestimated ED of nitrogen, presumably more important in high density containers. The increase in ash concentration at null density is very likely due to the mineral content of tap water used for feed preparation. ED of ash reached its maximum at 11.4 larvae/cm² but decreased with higher densities, indicating that the substrate accumulated minerals. This might be an experimental artefact: as BSFL used to start the trial were

not completely free from frass, high-density treatments were also inoculated with more frass than those of lower areal density, hence receiving more undigested ash from pre-experimental feeding. Higher frass ash content at high larval density could also result from a higher number of exuviae shed by BSFL, which bioaccumulate minerals in the cuticle in the form of calcium carbonate (Rubio *et al.*, 2022).

Life history traits values obtained were consistent with data from the literature on chicken feed: Liu *et al.* (2019) reported survival rates between 40 and 93% and Gold *et al.* (2018) summarised that prepupal FM could range between 99 and 184 mg. Larval DM content were in the range of 17.9 to 38.8% reported by Bosch *et al.* (2020). In the present study, high areal density caused stronger competition for feed resources, hence lower survival rate and individual FM and DM gain, as already observed (Brits, 2017; Jones *et al.*, 2019; Opore *et al.*, 2022). Dead BSFL could not be visually detected after seven days of feeding and were presumably consumed by other larvae or decomposed into the substrate. Similar adverse effects of density on survival or growth have been reported in other insect species such as *Drosophila melanogaster* (Henry *et al.*, 2020), *Musca domestica* (Kökdener and Kiper, 2020) or *Tenebrio molitor* (Deruytter and Coudron, 2022). Impaired larval DM content observed in the present work at high density suggests that limited access to feed caused BSFL to use their body energy resources by oxidising their fat reserve (Opore *et al.*, 2022). However, FCR (dry to fresh) was improved by increasing density down to an asymptote, indicating that BSFL performance as a group outweighed each individual underperformance. The same occurred with FCR (dry to dry) but only up to 11.4 larvae/cm². Interestingly, low densities ranging from 0.3 to 2.5 larvae/cm² are reported not to affect individual FM (Barragan-Fonseca *et al.*, 2018), presumably because the density was too low to induce competition. These contrasting findings suggest that, although increasing density results in competition and possible accumulation of toxic nitrogenous metabolic end products in the substrate like uric acid, urea or ammonia (Henry *et al.*, 2020; Weihrauch and O'Donnell, 2021), it also supports larval growth via mechanisms such as increased substrate temperature and aeration due to BSFL activity, cooperation through extra-oral digestion, or facilitated development of beneficial microbiota (Barragan-Fonseca *et al.*, 2018; Parra Paz *et al.*, 2015).

Differences between AD95 of ED of EE, ash, nitrogen, starch, energy and DM (6.7, 7.0, 7.1, 7.9, 8.1 and 9.0 larvae/cm², respectively) indicate that calculated optimal density to fully exploit the nutritional potential of a diet varies with the macronutrient considered. These differences could be related to the amount of each macronutrient in CF: it contained far less EE than starch and it might explain why EE requires fewer larvae to be optimally digested. Starch was the main component and the primary source of energy

in CF. Therefore, differences between AD95 of starch, DM and energy are relatively small. The difference with AD95 of FCR (dry to fresh) (8.5 larvae/cm²) is also low, which highlights that starch is an essential driver of BSFL weight gain (Barragan-Fonseca *et al.*, 2021).

Optimal areal density should be chosen based on the aim of the BSFL farming unit. Insect protein producers should be driven by the FCR (dry to dry), while for waste management stakeholders, insects are a tool for reducing organic waste and ED of DM should be the main driver (Myers *et al.*, 2008). According to present data, densities around 11 larvae/cm² (Figure 2D and 1A) would produce the best output for these two indicators. On the other hand, individual FM or DM gain is the most important indicator for insect producers looking for heavy BSFL (for reproduction purpose for example), and densities between 2 and 6 larvae/cm² (Figure 2E and 2F) should be used. These indications are consistent with broad recommendations available in the literature (Surendra *et al.*, 2020). However, optimal densities should also be based on economic considerations as BSFL neonates can be costly to produce. Insect producers must seek a trade-off between maximising the animal performance and minimising the cost of operations.

In the present study, ED had higher variability for densities below AD95 than for higher densities, which suggests asymptotic ED determination to be more reliable, repeatable and suitable for comparisons between types of feed than ED measured at lower densities. Therefore, when investigating the impact of a rearing parameter on ED, the maximal ED that can be obtained should be sought and not just one ED value for a given larval density. This asymptotic ED can be determined with a minimum of three different densities, for example 6, 12 and 18 larvae/cm², which would presumably result in ED below, around and beyond the asymptote respectively. The results of this study were obtained on CF with a 68% moisture content and in specific containers. Further investigations should focus on evaluating asymptotic ED and AD95 for different types of feed, but also for different feeding conditions likely to affect BSFL digestion, such as moisture content (Dzepe *et al.*, 2020), temperature (Harnden and Tomberlin, 2016), shape and size of the container (Miranda *et al.*, 2021), particle size (Brits, 2017) and addition of beneficial microbes (Zhang *et al.*, 2021). The latter represents an excellent opportunity to improve fibre digestion by using cellulolytic bacteria or fungi (Peguero *et al.*, 2022). ED of macronutrients of a standard diet could also be an insightful indicator when developing a genetic selection strategy for black soldier fly (Eriksson and Picard, 2021), as Zhou *et al.* (2013) showed that strain had an impact on ED of DM.

High-density containers reached a plateau of ED corresponding to the maximal amount of macronutrients

biologically available for BSFL and the substrate's microbial community. However, it remains unclear whether the remaining macronutrients were not digested because they were undigestible or because the substrate conditions evolved and inhibited further ingestion by the larvae. Indeed, at the end of the seven-day feeding period, frass in high-density containers was dry and powdery. This texture is inconsistent with BSFL mouthparts, adapted to ingest semi-liquid feed (Bruno *et al.*, 2020). Further investigations on the effect of water supplementation during growth on asymptotic ED are needed. Similarly, as frass was moist and sticky in low-density containers, BSFL might have been able to digest it further with an extended feeding period, leading to an ED closer to its maximal value. This hypothesis is supported by the ED of DM of 67.7% reported on BSFL fed CF with a density of 1.8 larvae/cm² after nine days (Gold *et al.*, 2020a), far above the value of 29.7% in seven days predicted by the model of the present study.

BSFL bioconversion process is more complex than digestion by monogastric animals or most other insects because BSFL live in their feeding substrate – a mixture of non-ingested feed, faeces, urine, dead individual and exuviae – which hosts an important microbial community that participates to the digestion (Figure 3). ED calculation is based on chemical analysis of distributed feed and frass and is a suitable indicator for insects living in their feed like BSFL. Yet more accurate digestibility indicators could be developed based on what is done in other insect or livestock species. Ingesta quantification could be fulfilled via the use of an undigestible tracer added to the feed and quantified in pure excreta, such as chromic oxide, allowing for Approximate Digestibility calculation (Equation 4)

(McGinnis and Kasting, 1964). Collecting pure excreta could either be done with an excreta collection device (Gold *et al.*, 2020b), by separating the top and bottom ends of BSFL by a perforated wall to allow them to feed from a compartment and release excreta in another one, or by putting clean BSFL in an empty container after a feeding period. However, all options lead to tiny excreta samples, are impractical on many insects, possibly stressful for the larvae with an impact on digestion, and could impair typical BSFL bioconversion processes (i.e. larval aggregation, microbial development, heat generation). Dissection of BSFL digestive tract to collect faeces should be avoided as the collected material may not be fully digested (Guillaume *et al.*, 1999). Approximate Digestibility has also been assessed in some insects via uric acid quantification. Dividing uric acid concentration in frass by uric acid concentration in pure excreta allows to determine the amount of excreta in frass, given that the feed does not contain uric acid (Bhattacharya and Waldbauer, 1969). Nevertheless, this method faces several limitations when applied to BSFL as its development also requires pure excreta collection, because some substrates distributed to BSFL such as poultry manure contain uric acid (Amanullah *et al.*, 2007) and finally because in moist and microbe-rich environment such as BSFL frass, uric acid would be quickly transformed in other nitrogenous compounds such as urea and ammonia (Henry *et al.*, 2020).

$$\text{Approximate Digestibility} = 1 - \frac{\text{Weight of excreta}}{\text{Weight of ingesta}} \quad (4)$$

$$\text{Apparent Digestibility} = 1 - \frac{\text{Weight of excreta} - \text{Weight of urines}}{\text{Weight of ingesta}} \quad (5)$$

$$\text{True Digestibility} = 1 - \frac{\text{Weight of excreta} - \text{Weight of urines} - \text{Weight of endogenous faeces}}{\text{Weight of ingesta}} \quad (6)$$

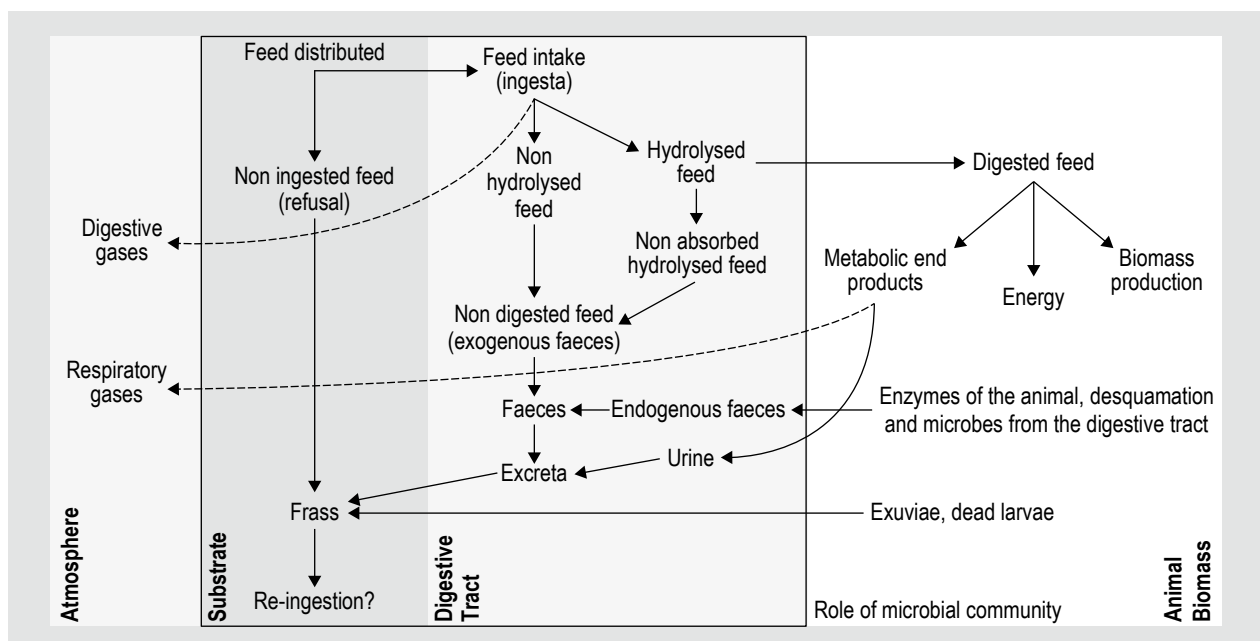


Figure 3. Complexity of breakdown processes of feed distributed to black soldier fly larvae during bioconversion.

In insects producing excreta that can be separated from other frass constituents, uric acid can also be quantified in excreta to calculate Apparent Digestibility (Equation 5), similarly to what is done in birds (Waldbauer, 1968). Yet the molecular form of nitrogenous metabolic end products excreted by Diptera remains unclear (Henry *et al.*, 2020) and further work is needed on urinary materials in frass and their degradation products, i.e. uric acid, allantoin, urea, ammonia, nitrite and nitrate (Barragan-Fonseca *et al.*, 2021). Materials produced in the digestive tract, also called ‘endogenous faeces’ (i.e. microbial biomass, enzymes and gut cells) (Figure 3), could also be quantified to calculate True Digestibility (Equation 6) (Barret *et al.*, 2005). However, this indicator was developed on animals producing an important amount of pure easy-to-collect faeces, such as poultry or pigs, and is complex to assess in small insects living in groups inside their feed. Estimated, approximate, apparent and true digestibilities (Equations 1, 4, 5 and 6, respectively) have close yet different definitions for BSFL and reflect both a rising precision in digestibility assessment and increasing technical issues. *In vitro* simulation of BSFL digestion could be promising but is limited by the understanding of the enzymatic and microbial activity of the digestive tract (Gold *et al.*, 2020c).

5. Conclusions

The present work proposes a method to evaluate BSFL feed digestion efficiency via asymptotic ED, a new indicator defined as the maximal fraction of initial macronutrients that are not in the frass at the end of the feeding period. ED of major macronutrients are assessed on a standard diet with a focus on the effect of larval areal density. Asymptotic ED corresponds to the maximal proportion of each macronutrient that BSFL and associated microbial community can potentially digest. Therefore, when formulating a diet for BSFL based on several ingredients, crude macronutrient content multiplied by the asymptotic ED of that macronutrient should be considered for each ingredient, rather than the sole crude macronutrient content. Further studies should focus on determining asymptotic ED of macronutrients in different types of ingredients that could be used as BSFL feed (such as agricultural side-streams, food waste or animal manure) and evaluate the effect of rearing conditions on asymptotic ED.

Author contributions

Conceptualisation: J. Guillaume, S. Mezdour, F. Marion-Poll, C. Terrol, P. Schmidely

Funding acquisition: J. Guillaume, S. Mezdour, F. Marion-Poll, C. Terrol, P. Schmidely

Supervision: S. Mezdour, F. Marion-Poll, C. Terrol, P. Schmidely

Trials: J. Guillaume

Data treatment: J. Guillaume, P. Schmidely

Writing original draft: J. Guillaume

Writing review and editing: J. Guillaume, F. Marion-Poll, C. Terrol, P. Schmidely

Acknowledgements

We would like to thank Agronutris and the French Ministry of Higher Education, Research and Innovation for their funding. We would also like to thank Béatrice Denis and Pénélope Tarapacki (EGCE) for their help in the set-up of the climate conditions, Ophélie Dhumez and Meriem Charef (AgroParisTech) for their support in samples analysis, and the R&D Team of Agronutris for their inputs on the experimental approach.

Funding

This work was funded by the private company Agronutris with support from the French Ministry of Higher Education, Research and Innovation in the frame of a CIFRE PhD grant attributed by the French National Association for Research and Technology. The company did not interfere with the conclusions of the study.

Conflict of interest

The authors declare no conflict of interest.

References

- Amanullah, M.M., Somasundaram, E., Vaiyapuri, K. and Sathyamoorthi, K., 2007. Poultry manure to crops: a review. *Agricultural Reviews- Agricultural Research Communications Centre India* 28: 216.
- Barragan-Fonseca, K.B., Dicke, M. and Van Loon, J.J.A., 2017. Nutritional value of the black soldier fly (*Hermetia illucens* L.) and its suitability as animal feed – a review. *Journal of Insects as Food and Feed* 3: 105-120. <https://doi.org/10.3920/jiff2016.0055>
- Barragan-Fonseca, K.B., Dicke, M. and Van Loon, J.J.A., 2018. Influence of larval density and dietary nutrient concentration on performance, body protein, and fat contents of black soldier fly larvae (*Hermetia illucens*). *Entomologia Experimentalis et Applicata* 166: 761-770. <https://doi.org/10.1111/eea.12716>
- Barragan-Fonseca, K.B., Gort, G., Dicke, M. and Van Loon, J.J.A., 2021. Nutritional plasticity of the black soldier fly (*Hermetia illucens*) in response to artificial diets varying in protein and carbohydrate concentrations. *Journal of Insects as Food and Feed* 7: 51-61. <https://doi.org/10.3920/jiff2020.0034>
- Barret, J.-P., Grosmond, G. and Simbélle, C., 2005. Alimentation des animaux domestiques, Zootechnie générale. Agriculture d'aujourd'hui. Lavoisier – Technique et Documentation, Paris, France, pp. 91-128.
- Bhattacharya, A.K. and Waldbauer, G.P., 1969. Faecal uric acid as an indicator in the determination of food utilization. *Journal of Insect Physiology* 15: 1129-1135. [https://doi.org/10.1016/0022-1910\(69\)90224-8](https://doi.org/10.1016/0022-1910(69)90224-8)

- Bosch, G. and Swanson, K.S., 2021. Effect of using insects as feed on animals: pet dogs and cats. *Journal of Insects as Food and Feed* 7: 795-805. <https://doi.org/10.3920/jiff2020.0084>
- Bosch, G., Oonincx, D.G.A.B., Jordan, H.R., Zhang, J., Van Loon, J.J.A., Van Huis, A. and Tomberlin, J.K., 2020. Standardisation of quantitative resource conversion studies with black soldier fly larvae. *Journal of Insects as Food and Feed* 6: 95-109. <https://doi.org/10.3920/jiff2019.0004>
- Brigode, C., Hobbi, P., Jafari, H., Verwilghen, F., Baeten, E. and Shavandi, A., 2020. Isolation and physicochemical properties of chitin polymer from insect farm side stream as a new source of renewable biopolymer. *Journal of Cleaner Production* 275: 122924. <https://doi.org/10.1016/j.jclepro.2020.122924>
- Brits, D., 2017. Improving feeding efficiencies of black soldier fly larvae, *Hermetia illucens* (L., 1758) (Diptera: Stratiomyidae: Hermetiinae) through manipulation of feeding conditions for industrial mass rearing. Master's Degree of AgriScience Thesis, Stellenbosch University, South Africa.
- Broeckx, L., Frooninckx, L., Slegers, L., Berrens, S., Noyens, I., Goossens, S., Verheyen, G., Wuyts, A. and Van Miert, S., 2021. Growth of black soldier fly larvae reared on organic side-streams. *Sustainability* 13: 12953. <https://doi.org/10.3390/su132312953>
- Bruno, D., Bonacci, T., Reguzzoni, M., Casartelli, M., Grimaldi, A., Tettamanti, G. and Brandmayr, P., 2020. An in-depth description of head morphology and mouthparts in larvae of the black soldier fly *Hermetia illucens*. *Arthropod Structure & Development* 58: 100969. <https://doi.org/10.1016/j.asd.2020.100969>
- Chan, Y.-K., Barraquio, W.L. and Knowles, R., 1994. N₂-fixing pseudomonads and related soil bacteria. *FEMS Microbiology Reviews* 13: 95-117. <https://doi.org/10.1111/j.1574-6976.1994.tb00037.x>
- Cifuentes, Y., Glaeser, S.P., Mvie, J., Bartz, J.O., Muller, A., Gutzeit, H.O., Vilcinskis, A. and Kampfer, P., 2020. The gut and feed residue microbiota changing during the rearing of *Hermetia illucens* larvae. *Antonie Van Leeuwenhoek* 113: 1323-1344. <https://doi.org/10.1007/s10482-020-01443-0>
- Cohn, Z., Latty, T. and Abbas, A., 2022. Understanding dietary carbohydrates in black soldier fly larvae treatment of organic waste in the circular economy. *Waste Management* 137: 9-19. <https://doi.org/10.1016/j.wasman.2021.10.013>
- Deruytter, D. and Coudron, C.L., 2022. The effects of density on the growth, survival and feed conversion of *Tenebrio molitor* larvae. *Journal of Insects as Food and Feed* 8: 141-146. <https://doi.org/10.3920/jiff2021.0057>
- Dörper, A., Veldkamp, T. and Dicke, M., 2021. Use of black soldier fly and house fly in feed to promote sustainable poultry production. *Journal of Insects as Food and Feed* 7: 761-780. <https://doi.org/10.3920/jiff2020.0064>
- Dzepe, D., Nana, P., Fotso, A., Tchuinkam, T. and Djouaka, R., 2020. Influence of larval density, substrate moisture content and feedstock ratio on life history traits of black soldier fly larvae. *Journal of Insects as Food and Feed* 6: 133-140. <https://doi.org/10.3920/jiff2019.0034>
- Eriksson, T. and Picard, C.J., 2021. Genetic and genomic selection in insects as food and feed. *Journal of Insects as Food and Feed* 7: 661-682. <https://doi.org/10.3920/jiff2020.0097>
- European Commission, 2021. Commission Regulation (EU) 2021/1925 of 5 November 2021 amending certain Annexes to Regulation (EU) No 142/2011 as regards the requirements for placing on the market of certain insect products and the adaptation of a containment method. *Official Journal of the European Union* L 393: 4-8. Available at: <http://data.europa.eu/eli/reg/2021/1925/oj>.
- Faisant, N., Planchot, V.r., Kozlowski, F., Pacouret, M.-P., Colonna, P. and Champ, M.M.-J., 1995. Resistant starch determination adapted to products containing high level of resistant starch. *Sciences des Aliments* 15: 83-89.
- Gobbi, P., Martínez-Sánchez, A. and Rojo, S., 2013. The effects of larval diet on adult life-history traits of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae). *European Journal of Endocrinology* 110: 461-468.
- Godin, B., Agneessens, R., Gofflot, S., Lamaudière, S., Sinnaeve, G., Gerin, P. and Delcarte, J., 2010. Review on analytical methods for lignocellulosic biomass structural polysaccharides. *Biotechnology, Agronomy, Society and Environment* 15: 165-182.
- Gold, M., Binggeli, M., Kurt, F., de Wouters, T., Reichlin, M., Zurbrugg, C., Mathys, A. and Kreuzer, M., 2020b. Novel experimental methods for the investigation of *Hermetia illucens* (Diptera: Stratiomyidae) larvae. *Journal of Insect Science* 20: 21. <https://doi.org/10.1093/jisesa/ieaa057>
- Gold, M., Cassar, C.M., Zurbrugg, C., Kreuzer, M., Boulos, S., Diener, S. and Mathys, A., 2020a. Biowaste treatment with black soldier fly larvae: increasing performance through the formulation of biowastes based on protein and carbohydrates. *Waste Management* 102: 319-329. <https://doi.org/10.1016/j.wasman.2019.10.036>
- Gold, M., Egger, J., Scheidegger, A., Zurbrugg, C., Bruno, D., Bonelli, M., Tettamanti, G., Casartelli, M., Schmitt, E., Kerkaert, B., Smet, J., Van Campenhout, L. and Mathys, A., 2020c. Estimating black soldier fly larvae biowaste conversion performance by simulation of midgut digestion. *Waste Management* 112: 40-51. <https://doi.org/10.1016/j.wasman.2020.05.026>
- Gold, M., Tomberlin, J.K., Diener, S., Zurbrugg, C. and Mathys, A., 2018. Decomposition of biowaste macronutrients, microbes, and chemicals in black soldier fly larval treatment: a review. *Waste Management* 82: 302-318. <https://doi.org/10.1016/j.wasman.2018.10.022>
- Greenwood, M.P., Hull, K.L., Brink-Hull, M., Lloyd, M. and Rhode, C., 2021. Feed and host genetics drive microbiome diversity with resultant consequences for production traits in mass-reared black soldier fly (*Hermetia illucens*) larvae. *Insects* 12: 1082. <https://doi.org/10.3390/insects12121082>
- Guillaume, J., Kaushik, S., Bergot, P. and Métailler, R., 1999. Nutrition et alimentation des poissons et crustacés. INRA / Ifremer, 490 pp.
- Harnden, L.M. and Tomberlin, J.K., 2016. Effects of temperature and diet on black soldier fly, *Hermetia illucens* (L.) (Diptera: Stratiomyidae), development. *Forensic Science International* 266: 109-116. <https://doi.org/10.1016/j.forsciint.2016.05.007>
- Henry, Y., Tarapacki, P. and Colinet, H., 2020. Larval density affects phenotype and surrounding bacterial community without altering gut microbiota in *Drosophila melanogaster*. *FEMS Microbiology Ecology* 96: fiae055. <https://doi.org/10.1093/femsec/fiae055>

- Hopkins, I., Newman, L.P., Gill, H. and Danaher, J., 2021. The influence of food waste rearing substrates on black soldier fly larvae protein composition: a systematic review. *Insects* 12: 608. <https://doi.org/10.3390/insects12070608>
- Intayung, D., Chundang, P., Srikachar, S. and Kovitvadhi, A., 2021. Ontogenic development of the digestive enzymes and chemical composition of *Hermetia illucens* larvae of different ages. *Entomologia Experimentalis et Applicata* 169: 665-673. <https://doi.org/10.1111/eea.13063>
- Jones, B.M., Tomberlin, J.K. and Stull, V., 2019. Impact of larval competition on life-history traits of the black soldier fly (Diptera: Stratiomyidae). *Annals of the Entomological Society of America* 112: 505-510. <https://doi.org/10.1093/aesa/saz014>
- Klammsteiner, T., Walter, A., Bogataj, T., Heussler, C.D., Stres, B., Steiner, F.M., Schlick-Steiner, B.C. and Insam, H., 2021. Impact of processed food (canteen and oil wastes) on the development of black soldier fly (*Hermetia illucens*) larvae and their gut microbiome functions. *Frontiers in Microbiology* 12: 619112. <https://doi.org/10.3389/fmicb.2021.619112>
- Kökdener, M. and Kiper, F., 2020. Effects of larval population density and food type on the life cycle of *Musca domestica* (Diptera: Muscidae). *Environmental Entomology* 50: 324-329. <https://doi.org/10.1093/ee/nvaa165>
- Lalander, C., Diener, S., Zurbrügg, C. and Vinnerås, B., 2019. Effects of feedstock on larval development and process efficiency in waste treatment with black soldier fly (*Hermetia illucens*). *Journal of Cleaner Production* 208: 211-219. <https://doi.org/10.1016/j.jclepro.2018.10.017>
- Liu, C., Wang, C. and Yao, H., 2019. Comprehensive resource utilization of waste using the black soldier fly (*Hermetia illucens* (L.)) (Diptera: Stratiomyidae). *Animals* 9: 349. <https://doi.org/10.3390/ani9060349>
- McGinnis, A.J. and Kasting, R., 1964. Comparison of gravimetric and chromic oxide methods for measuring percentage utilization and consumption of food by phytophagous insects. *Journal of Insect Physiology* 10: 989-995. [https://doi.org/10.1016/0022-1910\(64\)90086-1](https://doi.org/10.1016/0022-1910(64)90086-1)
- Miranda, C.D., Crippen, T.L., Cammack, J.A. and Tomberlin, J.K., 2021. Black soldier fly, *Hermetia illucens* (L.) (Diptera: Stratiomyidae), and house fly, *Musca domestica* L. (Diptera: Muscidae), larvae reduce livestock manure and possibly associated nutrients: an assessment at two scales. *Environmental Pollution* 282: 116976. <https://doi.org/10.1016/j.envpol.2021.116976>
- Myers, H.M., Tomberlin, J.K., Lambert, B.D. and Kattes, D., 2008. Development of black soldier fly (Diptera: Stratiomyidae) larvae fed dairy manure. *Environmental Entomology* 37: 11-15. [https://doi.org/10.1603/0046-225x\(2008\)37\[11:Dobsfd\]2.0.Co;2](https://doi.org/10.1603/0046-225x(2008)37[11:Dobsfd]2.0.Co;2)
- Opare, L.O., Holm, S. and Esperk, T., 2022. Temperature-modified density effects in the black soldier fly: low larval density leads to large size, short development time and high fat content. *Journal of Insects as Food and Feed* 8: 783-802. <https://doi.org/10.3920/jiff2021.0147>
- Parodi, A., De Boer, I.J.M., Gerrits, W.J.J., Van Loon, J.J.A., Heetkamp, M.J.W., Van Schelt, J., Bolhuis, J.E. and Van Zanten, H.H.E., 2020. Bioconversion efficiencies, greenhouse gas and ammonia emissions during black soldier fly rearing – a mass balance approach. *Journal of Cleaner Production* 271: 122488. <https://doi.org/10.1016/j.jclepro.2020.122488>
- Parodi, A., Gerrits, W.J.J., Van Loon, J.J.A., De Boer, I.J.M., Aarnink, A.J.A. and Van Zanten, H.H.E., 2021. Black soldier fly reared on pig manure: bioconversion efficiencies, nutrients in the residual material, greenhouse gas and ammonia emissions. *Waste Management* 126: 674-683. <https://doi.org/10.1016/j.wasman.2021.04.001>
- Parra Paz, A.S., Carrejo, N.S. and Gómez Rodríguez, C.H., 2015. Effects of larval density and feeding rates on the bioconversion of vegetable waste using black soldier fly larvae *Hermetia illucens* (L.) (Diptera: Stratiomyidae). *Waste and Biomass Valorization* 6: 1059-1065. <https://doi.org/10.1007/s12649-015-9418-8>
- Peguero, D.A., Gold, M., Vandeweyer, D., Zurbrügg, C. and Mathys, A., 2022. A review of pretreatment methods to improve agri-food waste bioconversion by black soldier fly larvae. *Frontiers in Sustainable Food Systems* 5: 745894. <https://doi.org/10.3389/fsufs.2021.745894>
- Permana, A.D. and Ramadhani Eka Putra, J.E.N., 2018. Growth of black soldier fly (*Hermetia illucens*) larvae fed on spent coffee ground. *IOP Conference Series: Earth and Environmental Science* 187: 012070. <https://doi.org/10.1088/1755-1315/187/1/012070>
- R Core Team, 2021. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. Available at: <https://www.R-project.org/>.
- Rehman, K.u., Rehman, A., Cai, M., Zheng, L., Xiao, X., Somroo, A.A., Wang, H., Li, W., Yu, Z. and Zhang, J., 2017. Conversion of mixtures of dairy manure and soybean curd residue by black soldier fly larvae (*Hermetia illucens* L.). *Journal of Cleaner Production* 154: 366-373. <https://doi.org/10.1016/j.jclepro.2017.04.019>
- Rehman, K.U., Ur Rehman, R., Somroo, A.A., Cai, M., Zheng, L., Xiao, X., Ur Rehman, A., Rehman, A., Tomberlin, J.K., Yu, Z. and Zhang, J., 2019. Enhanced bioconversion of dairy and chicken manure by the interaction of exogenous bacteria and black soldier fly larvae. *Journal of Environmental Management* 237: 75-83. <https://doi.org/10.1016/j.jenvman.2019.02.048>
- Rubio, P., Todolí, J.L., Martínez-Sánchez, A. and Rojo, S., 2022. Evolution of the mineral concentration and bioaccumulation of the black soldier fly, *Hermetia illucens*, feeding on two different larval media. *Journal of Insects as Food and Feed* 8: 367-378. <https://doi.org/10.3920/jiff2021.0068>
- Sauvant, D., Perez, J.-M. and Tran, G., 2002. Tables de composition et de valeur nutritive des matières premières destinées aux animaux d'élevage. INRA, Paris, France, 301 pp.
- Schreven, S.J.J., de Vries, H., Hermes, D.A.G., Smidt, H., Dicke, M. and van Loon, J.J., 2021. Relative contributions of egg-associated and substrate-associated microorganisms to black soldier fly larval performance and microbiota. *FEMS Microbiology Ecology* 97: fiab054. <https://doi.org/10.1093/femsec/fiab054>

- Schreven, S.J.J., De Vries, H., Hermes, G.D.A., Zeni, G., Smidt, H., Dicke, M. and Van Loon, J.J.A., 2022. Black soldier fly larvae influence internal and substrate bacterial community composition depending on substrate type and larval density. *Applied and Environmental Microbiology* 88: e00084-00022. <https://doi.org/10.1128/aem.00084-22>
- Seyedalmoosavi, M.M., Mielenz, M., Veldkamp, T., Das, G. and Metges, C.C., 2022. Growth efficiency, intestinal biology, and nutrient utilization and requirements of black soldier fly (*Hermetia illucens*) larvae compared to monogastric livestock species: a review. *Journal of Animal Science and Biotechnology* 13: 31. <https://doi.org/10.1186/s40104-022-00682-7>
- Stein, H.H., Lagos, L.V. and Casas, G.A., 2016. Nutritional value of feed ingredients of plant origin fed to pigs. *Animal Feed Science and Technology* 218: 33-69. <https://doi.org/10.1016/j.anifeedsci.2016.05.003>
- Stelmock, R.L., Husby, F.M. and Brundage, A.L., 1985. Application of Van Soest Acid detergent fiber method for analysis of shellfish chitin1. *Journal of Dairy Science* 68: 1502-1506. [https://doi.org/10.3168/jds.S0022-0302\(85\)80989-9](https://doi.org/10.3168/jds.S0022-0302(85)80989-9)
- Sun, Q.X., Li, X.C., Tan, X.H., Dong, Y.W., You, C.H., Yuan, H.L., Lu, Y.Y., Huang, Y.H. and Zhou, M., 2022. Digestive physiological characteristics of black soldier fly larvae reared on five organic wastes. *Journal of Insects as Food and Feed* 8: 451-467. <https://doi.org/10.3920/jiff2021.0059>
- Surendra, K.C., Tomberlin, J.K., van Huis, A., Cammack, J.A., Heckmann, L.L. and Khanal, S.K., 2020. Rethinking organic wastes bioconversion: Evaluating the potential of the black soldier fly (*Hermetia illucens* (L.)) (Diptera: Stratiomyidae) (BSF). *Waste Management* 117: 58-80. <https://doi.org/10.1016/j.wasman.2020.07.050>
- Vaissaire, J.P., 2018. *Mémento de zootechnie*. Editions France Agricole, Paris, 338 pp.
- Waldbauer, G.P., 1968. The consumption and utilization of food by insects, advances in insect physiology Volume 5. In: Beament, J.W.L., Treherne, J.E. and Wigglesworth, V.B. (eds) *Advances in insect physiology*, Academic Press Inc., London, United Kingdom, pp. 229-288. [https://doi.org/10.1016/s0065-2806\(08\)60230-1](https://doi.org/10.1016/s0065-2806(08)60230-1)
- Weihrauch, D. and O'Donnell, M.J., 2021. Mechanisms of nitrogen excretion in insects. *Current Opinions in Insect Science* 47: 25-30. <https://doi.org/10.1016/j.cois.2021.02.007>
- Zaefarian, F., Abdollahi, M.R. and Ravindran, V., 2015. Starch digestion in broiler chickens fed cereal diets. *Animal Feed Science and Technology* 209: 16-29. <https://doi.org/10.1016/j.anifeedsci.2015.07.020>
- Zhang, J.B., Yu, Y.Q., Tomberlin, J.K., Cai, M.M., Zheng, L.Y. and Yu, Z.N., 2021. Organic side streams: using microbes to make substrates more fit for mass producing insects for use as feed. *Journal of Insects as Food and Feed* 7: 597-604. <https://doi.org/10.3920/jiff2020.0078>
- Zhou, F., Tomberlin, J.K., Zheng, L., Yu, Z. and Zhang, J., 2013. Developmental and waste reduction plasticity of three black soldier fly strains (Diptera: Stratiomyidae) raised on different livestock manures. *Journal of Medical Entomology* 50: 1224-1230. <https://doi.org/10.1603/me13021>